THE CLASS OF ARROWS MADE INVERTIBLE BY AN EXACT FUNCTOR IS A SATURATED MULTIPLICATIVE SYSTEM

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 1488-1522, following the statement at 1473-1486.

Editorial note. MS6 is the TR3 axiom output combined with lemma-third-isomorphism-triangle; both pieces are named and no additional glue is hidden.

Editorial note. RMS3 marked dual by the author, following the established dual-mirror counting practice.

Allowed standard prerequisites (editorial).

- Formal multiplicative system consequences. lemma-localization-conditions (same chapter, cited at ~1494). Yields MS2 from the remaining axioms.
- Third entry of a morphism of triangles. lemma-third-isomorphism-triangle (same chapter, cited at ~1490) and lemma-third-object-zero (same chapter, cited at ~1512). MS6 completion and the vanishing F(Q) = 0 behind LMS3.
- Representable functor is homological. lemma-representable-homological (same chapter, cited at ~1506). Lifts a = f - g through the triangle to i: Q -> Y with i d = a.

Source reference. Original reference 04VC: focused-seeds/geometry/S9-der-triangle-functor-localize/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 05Q8: focused-seeds/geometry/S9-der-triangle-functor-localize/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original definition 05R2: sources/stacks-derived/pinned-original-source.tex, source lines 1374–1374. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 05R3: sources/stacks-derived/pinned-original-source.tex, source lines 1405–1405. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Date: stacks-project:derived.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 0149: sources/stacks-derived/pinned-original-source.tex, source lines 603–603. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 014A: sources/stacks-derived/pinned-original-source.tex, source lines 631–631. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 05QR: sources/stacks-derived/pinned-original-source.tex, source lines 833–833. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let $F : \mathcal{D} \to \mathcal{D}'$ be an exact functor of pre-triangulated categories. Let*

$$S = \{f \in \mathit{Arrows}(\mathcal{D}) \mid F(f) \text{ is an isomorphism}\}$$

Then S is a saturated (see Categories, Definition 05Q8) multiplicative system compatible with the triangulated structure on $\mathcal{D}$.

Proof. We have to prove axioms MS1 – MS6, see Categories, Definitions 04VC and 05Q8 and Definition 05R2. MS1, MS4, and MS5 are direct from the definitions. MS6 follows from TR3 and Lemma 014A. By Lemma 05R3 we conclude that MS2 holds. To finish the proof we have to show that MS3 holds. To do this let $f, g : X \to Y$ be morphisms of $\mathcal{D}$, and let $t : Z \to X$ be an element of S such that $f \circ t = g \circ t$. As $\mathcal{D}$ is additive this simply means that $a \circ t = 0$ with $a = f - g$. Choose a distinguished triangle (Z, X, Q, t, d, h) using TR1. Since $a \circ t = 0$ we see by Lemma 0149 there exists a morphism $i : Q \to Y$ such that $i \circ d = a$. Finally, using TR1 again we can choose a triangle (Q, Y, W, i, j, k). Here is a picture

$$\begin{array}{ccccccc} Z & \xrightarrow[t]{} & X & \xrightarrow[d]{} & Q & \longrightarrow & Z[1] \\ & & \downarrow{\scriptstyle 1} & & \downarrow{\scriptstyle i} & & \\ & & X & \xrightarrow[a]{} & Y & & \\ & & & & \downarrow{\scriptstyle j} & & \\ & & & & W & & \end{array}$$

OK, and now we apply the functor F to this diagram. Since $t \in S$ we see that $F(Q) = 0$, see Lemma 05QR. Hence $F(j)$ is an isomorphism by the same lemma, i.e., $j \in S$. Finally, $j \circ a = j \circ i \circ d = 0$ as $j \circ i = 0$. Thus $j \circ f = j \circ g$ and we see that LMS3 holds. The proof of RMS3 is dual. □